

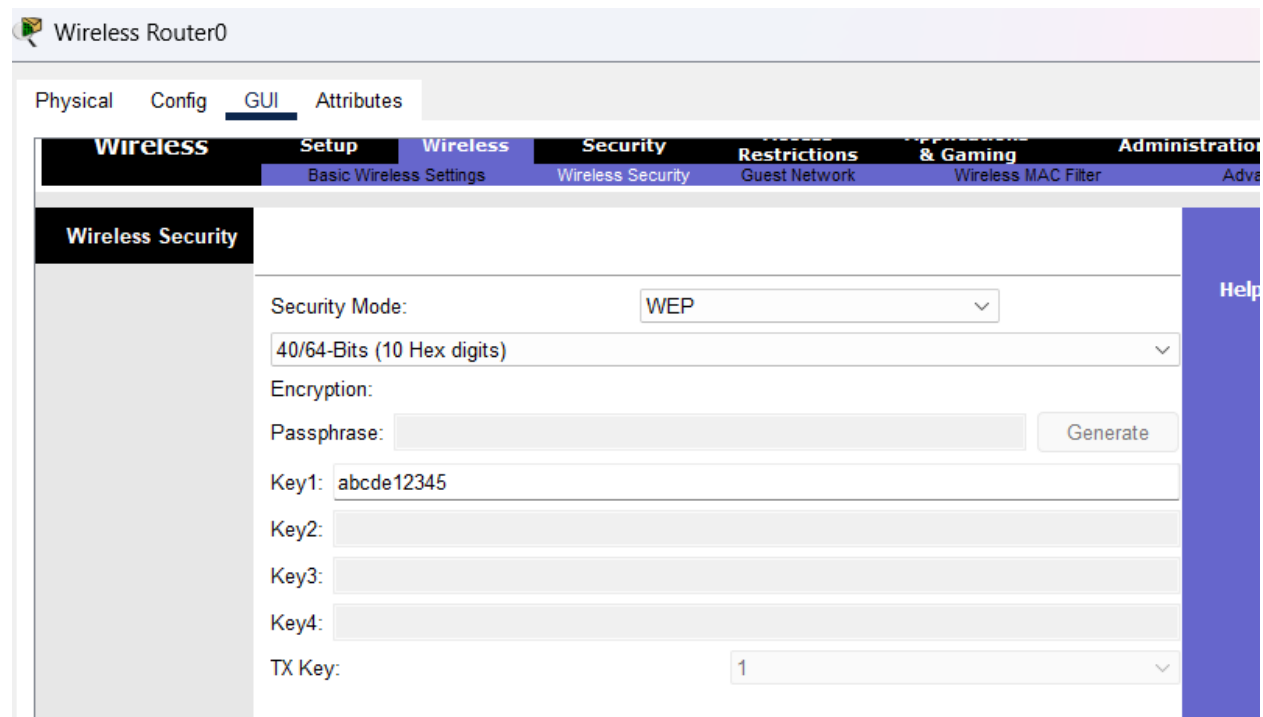
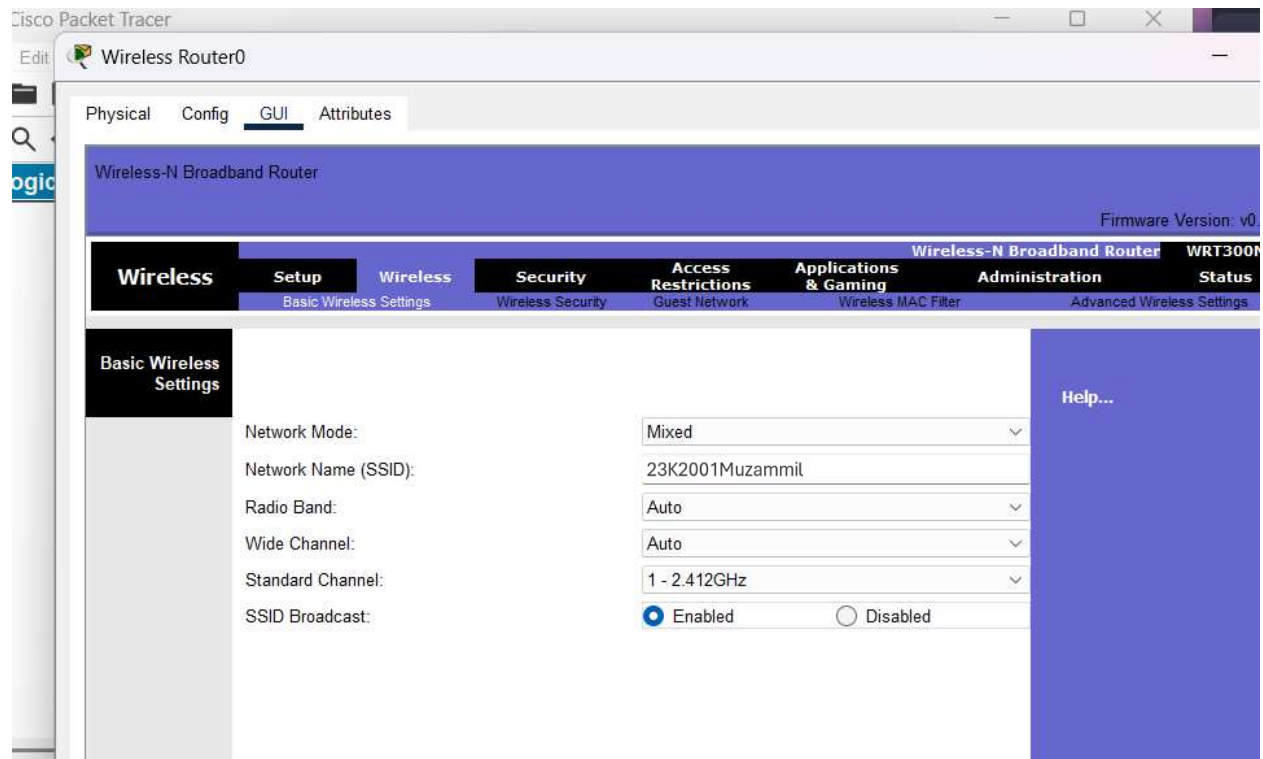
Computer Networks

LAB#08



23K-2001

BCS-5J



Link Information

Connect

Profiles

Below is a list of available wireless networks. To search for more wireless networks, click the **Refresh** button. To view more information about a network, select the wireless network name. To connect to that network, click the **Connect** button below.

Wireless Network Name	CH	Signal
23K2001Muzammil	1	100%

Site Information

Wireless Mode Infrastructure
Network Type Mixed B/G/N
Radio Band Auto
Security WEP
MAC Address 0001.4302.8806

Refresh

Connect

2.4GHz



Adapter is Inactive

Smartphone1

PhysicalConfigDesktopProgrammingAttributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

Wireless0

3G/4G Cell1

Bluetooth

Wireless0

Port Status

On

Bandwidth

300 Mbps

MAC Address

00E0.A32A.1B16

SSID

23K2001Muzammil

Authentication

☐ Disabled

☒ WEP

☐ WPA-PSK

☐ WPA

☐ 802.1X

☐ WPA2-PSK

☐ WPA2

Method:

WEP Key

abcde12345

PSK Pass Phrase

User ID

Password

MD5

User Name

Password

Encryption Type

40/64-Bits (10 Hex digits)

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.0.59

Subnet Mask

255.255.255.0

IPv6 Configuration

☒ Automatic

☐ Static

IPv6 Address

/

Link Local Address

FE80::2E0:A3FF:FE2A:1B16

Tablet PC0

PhysicalConfigDesktopProgrammingAttributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

Wireless0

3G/4G Cell1

Bluetooth

Wireless0

Port Status

On

Bandwidth

300 Mbps

MAC Address

00D0.5803.B3CE

SSID

23K2001Muzammil

Authentication

☐ Disabled

☒ WEP

☐ WPA-PSK

☐ WPA

☐ 802.1X

☒ WPA2-PSK

☐ WPA2

Method:

WEP Key

abcde12345

PSK Pass Phrase

User ID

Password

MD5

User Name

Password

Encryption Type

40/64-Bits (10 Hex digits)

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.0.60

Subnet Mask

255.255.255.0

IPv6 Configuration

☒ Automatic

☐ Static

IPv6 Address

/

Link Local Address

FE80::2D0:58FF:FE03:B3CE

WEP Key Needed for Connection

This wireless network has WEP encryption enabled. To connect to this network, select the level of WEP encryption. Enter the required passphrase or WEP key in the appropriate field below. Then click the **Connect**.

Security WEP

Please select the wireless security method used by your existing wireless network.

WEP 64-bit

To use WEP encryption, select 64-bit or 128-bit

Passphrase

The Passphrase is case-sensitive and should be no more than 16 characters in length.

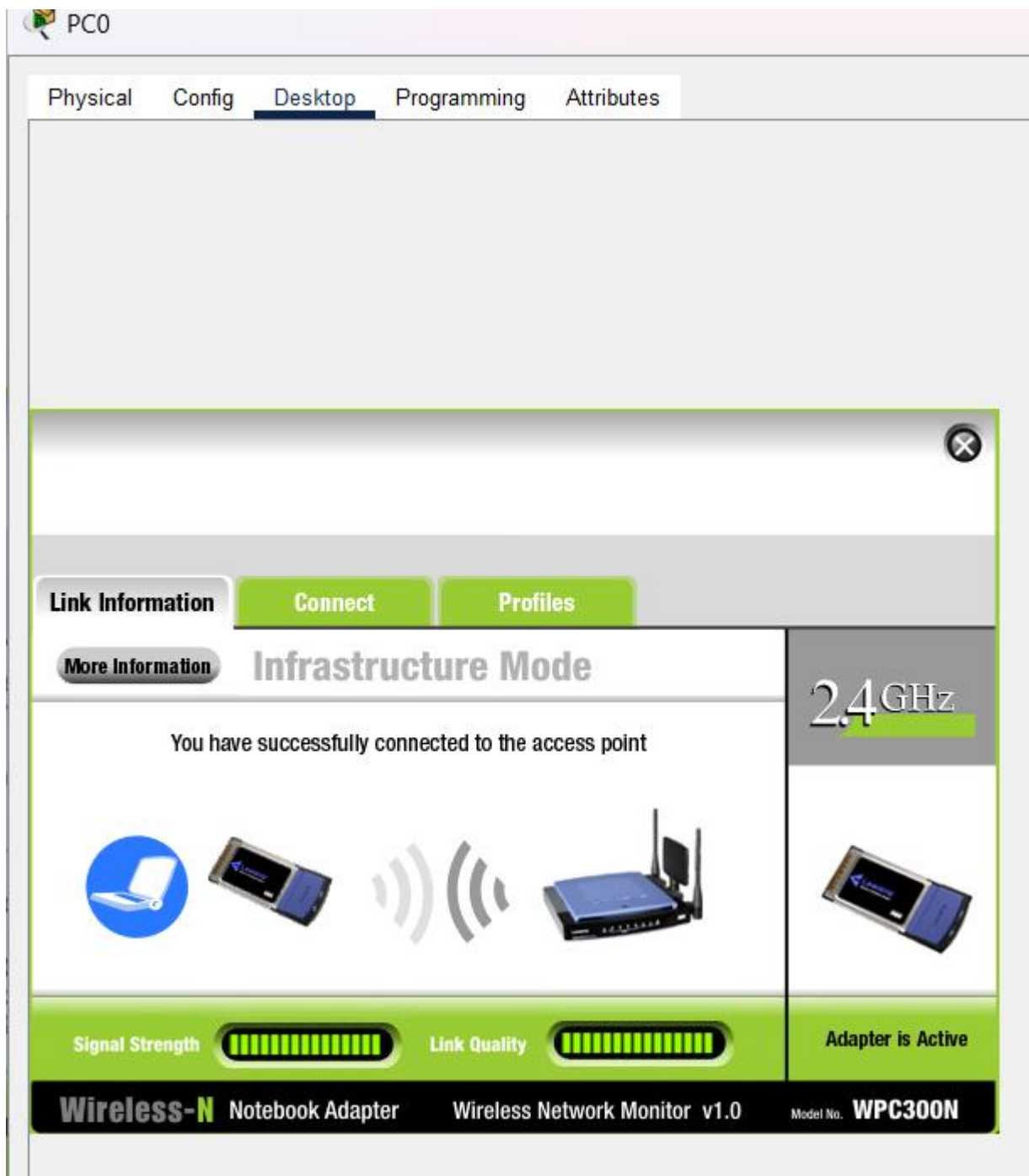
WEP Key 1 abcde12345

When entering this manually, it should be 10 characters for 64-bit encryption or 26 characters for 128-bit encryption. Valid hexadecimal characters are "A" through "F" and numbers "0" through "9".

Cancel

Connect

Active



```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
```



```

Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#line vty 0 15
Router(config-line)#password abc
Router(config-line)#login
Router(config-line)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#enable password abc
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

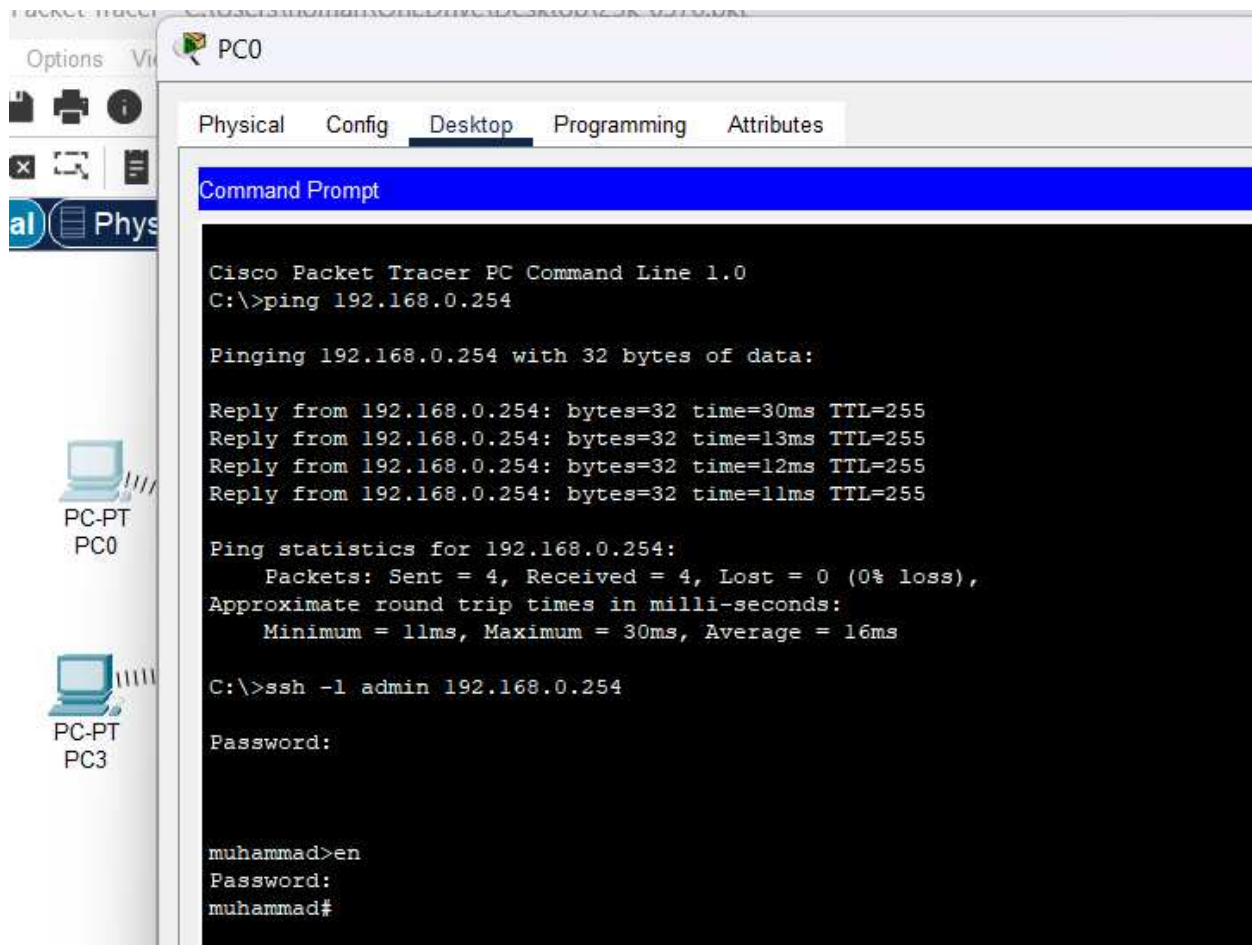
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname
% Incomplete command.
Router(config)#hostname muhammad
muhammad(config)#ip domain name muzammil
muhammad(config)#crypto key generate rsa
The name for the keys will be: muhammad.muzammil
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

muhammad(config)#ip ssh version 2
*Mar 1 0:23:58.956: %SSH-5-ENABLED: SSH 1.99 has been enabled
muhammad(config)#transport input ssh
^
% Invalid input detected at '^' marker.

muhammad(config)#line vty 0 15
muhammad(config-line)#transport input ssh
muhammad(config-line)#

```

Server0

PhysicalConfigServicesDesktopProgrammingAttributes

IP Configuration

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.0.62

Subnet Mask

255.255.255.0

Default Gateway

192.168.0.1

DNS Server

0.0.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

Link Local Address

FE80::260:47FF:FE28:1EA

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

PC4

Physical

Config

Desktop

Programming

Attributes

IP Configuration

X

Interface

FastEthernet0

IP Configuration

DHCP

Static

IPv4 Address

192.168.0.61

Subnet Mask

255.255.255.0

Default Gateway

192.168.0.1

DNS Server

192.168.0.62

IPv6 Configuration

Automatic

Static

IPv6 Address

/

Link Local Address

FE80::201:43FF:FE2D:EE37

Default Gateway

DNS Server

802.1X

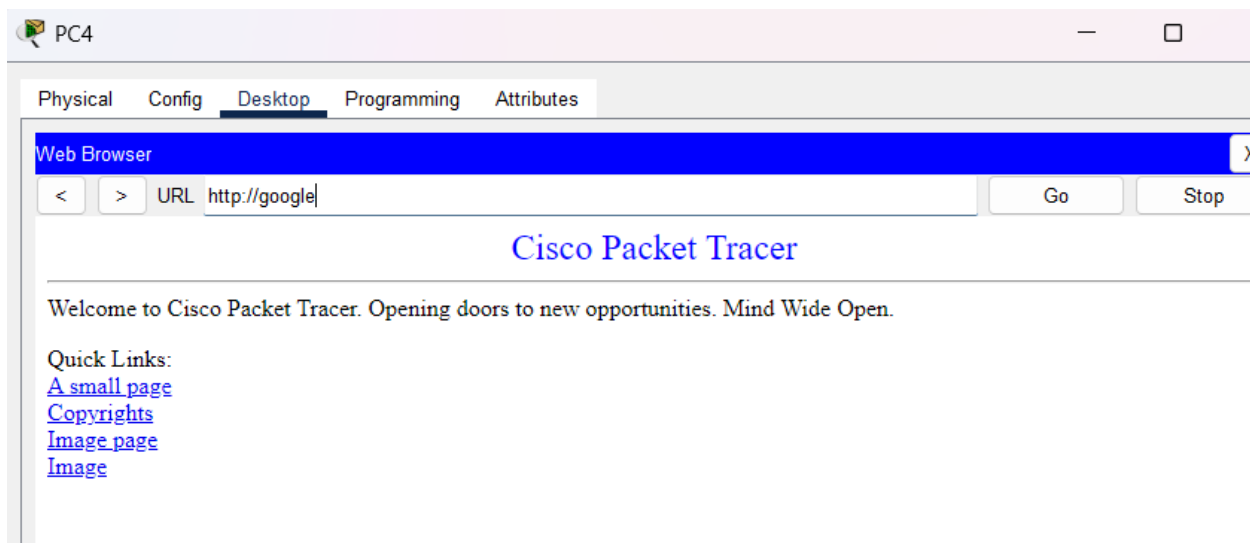
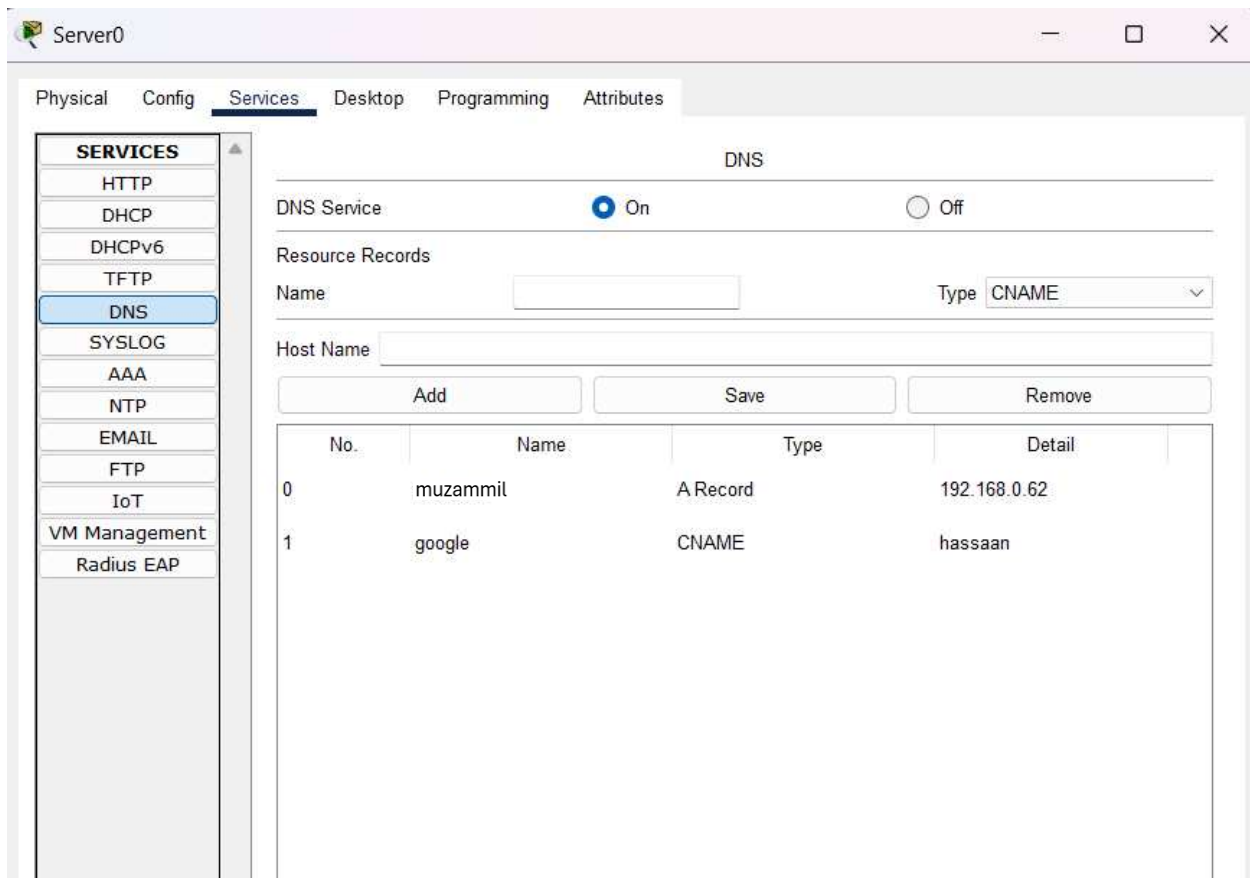
Use 802.1X Security

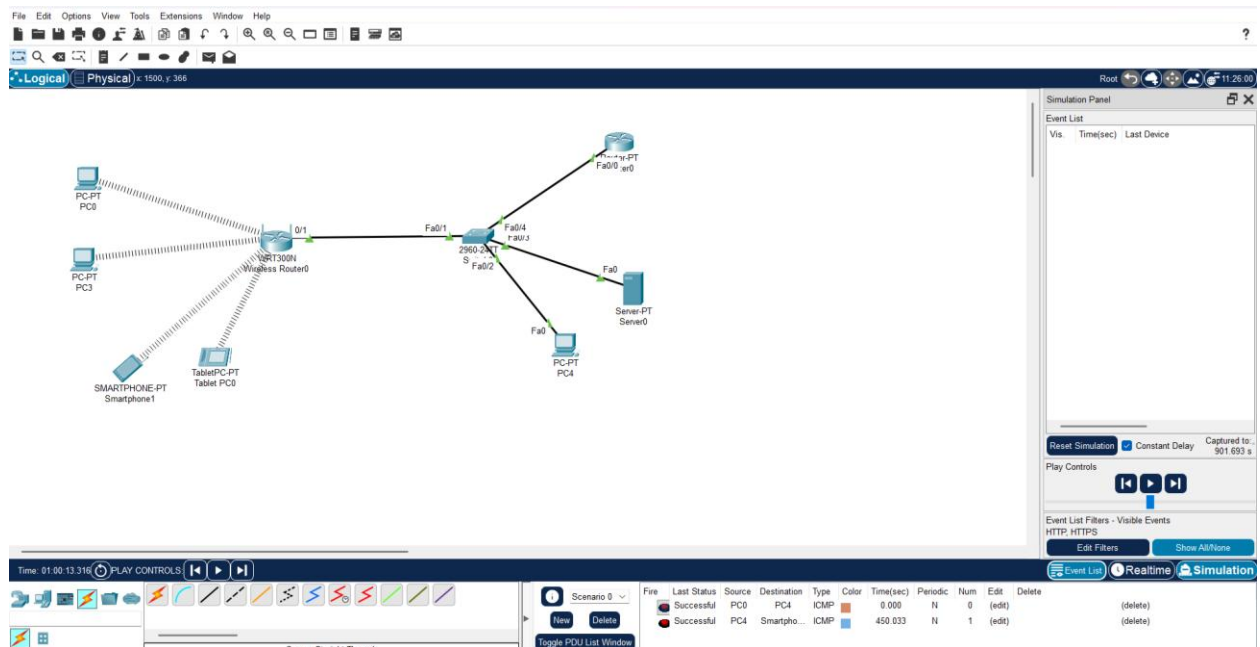
Authentication

MD5

Username

Password





Server0

PhysicalConfigServicesDesktopProgrammingAttributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

EMAIL

SMTP Service

☒ ON☐ OFF

POP3 Service

☒ ON☐ OFF

Domain Name: hotmail.com

Set

User Setup

User

Password

muzammil

muhammad

+

-

Change

Password

PC0

Physical Config **Desktop** Programming Attributes

Configure Mail X

User Information

Your Name: muzammil

Email Address: muzammil@hotmail.com

Server Information

Incoming Mail Server: 192.168.0.62

Outgoing Mail Server: 192.168.0.62

Logon Information

User Name: muzammil

Password: ●●●

Save Remove Clear Reset

PC4

Physical Config **Desktop** Programming Attributes

Configure Mail X

User Information

Your Name: muhammad

Email Address: muhammad@hotmail.com

Server Information

Incoming Mail Server: 192.168.0.62

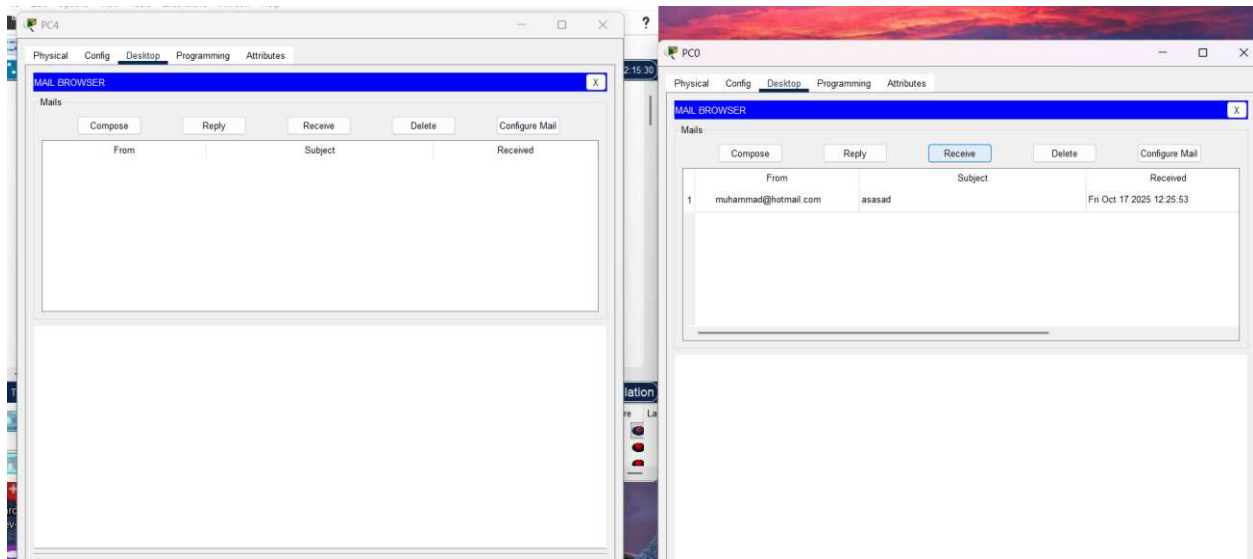
Outgoing Mail Server: 192.168.0.62

Logon Information

User Name: muhammad

Password: ●●●

Save Remove Clear Reset



Q2

```
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed
state to up

Router con0 is now available

Press RETURN to get started.

Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fal/0
Router(config-if)#ip address 5.7.1.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#
```

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 192.168.57.101 255.255.255.0
Router(config-if)#no shutdown
```

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 6.0.0.0 255.0.0.0 5.7.1.2
```

```
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip nat translation
Pro  Inside global      Inside local      Outside local      Outside global
---  5.7.1.1              192.168.57.104   ---               ---
```

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed  
Router(config-if)#%IP-4-DUPADDR: Duplicate address 192.168.58.101 on FastEthernet0/0:0002.4AD6.8987
```

```
Router con0 is now available
```

```
Press RETURN to get started.
```

```
Router>en  
Router#config t  
Enter configuration commands, one per line. End with CNTL/Z.  
Router(config)#int fa1/0  
Router(config-if)#ip address 5.7.1.2 255.255.255.252  
Router(config-if)#no shutdown
```

```
Router(config-if)#exit
Router(config)#int fa0/0
Router(config-if)#ip address 192.168.58.101
% Incomplete command.
Router(config-if)#ip address 192.168.58.101 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
```

 Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
170 West Tasman Drive
San Jose, California 95134-1706

Cisco Internetwork Operating System Software
IOS (tm) PT1000 Software (PT1000-I-M), Version 12.2(28), RELEASE SOFTWARE (fc5)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2005 by cisco Systems, Inc.
Compiled Wed 27-Apr-04 19:01 by miwang

PT 1001 (PTSC2005) processor (revision 0x200) with 60416K/5120K bytes of memory
.
Processor board ID PT0123 (0123)
PT2005 processor: part number 0, mask 01
Bridging software.
X.25 software, Version 3.0.0.
4 FastEthernet/IEEE 802.3 interface(s)
2 Low-speed serial(sync/async) network interface(s)
32K bytes of non-volatile configuration memory.
63488K bytes of ATA CompactFlash (Read/Write)

Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip nat inside
Router(config-if)#exit
Router(config)#int fa1/0
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#ip nat inside source static 192.168.57.104 5.7.1.1
Router(config)#ip route 192.168.58.0 255.255.255.0 5.7.1.2
Router(config)#
```

Copy F

```

Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip nat inside
Router(config-if)#exit
Router(config)#int fal/0
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#access-list 1 permit 192.168.58.0 0.0.0.255
Router(config)#ip nat pool dyn_pool 6.0.0.1 6.0.0.10 netmask 255.0.0.0
Router(config)#ip nat inside source list 1 pool dyn_pool
Router(config)#ip route 192.168.57.0 255.255.255.0 5.7.1.1
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip nat translations
Router#show ip nat statistics
Total translations: 0 (0 static, 0 dynamic, 0 extended)
Outside Interfaces: FastEthernet1/0
Inside Interfaces: FastEthernet0/0
Hits: 0 Misses: 0
Expired translations: 0
Dynamic mappings:
-- Inside Source
access-list 1 pool dyn_pool refCount 0
  pool dyn_pool: netmask 255.0.0.0
    start 6.0.0.1 end 6.0.0.10
    type generic, total addresses 10 , allocated 0 (0%), misses 0

Router#show ip nat translations

```

Pro	Inside global	Inside local	Outside local	Outside global
icmp	6.0.0.1:1	192.168.58.103:1	192.168.57.103:1	192.168.57.103:1
icmp	6.0.0.1:2	192.168.58.103:2	192.168.57.103:2	192.168.57.103:2
icmp	6.0.0.1:3	192.168.58.103:3	192.168.57.103:3	192.168.57.103:3
icmp	6.0.0.1:4	192.168.58.103:4	192.168.57.103:4	192.168.57.103:4

PC1

Physical

Config

Desktop

Programming

Attributes

IP Configuration

X

Interface

FastEthernet0

▼

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.57.103

Subnet Mask

255.255.255.0

Default Gateway

192.168.57.101

DNS Server

0.0.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

/

Link Local Address

FE80::207:ECFF:FE04:37C1

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

MD5

▼

Username

Password

PC2

PhysicalConfigDesktopProgrammingAttributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.57.102

Subnet Mask

255.255.255.0

Default Gateway

192.168.57.101

DNS Server

0.0.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

/

Link Local Address

FE80::201:63FF:FE01:8D4A

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

PC5

Physical

Config

Desktop

Programming

Attributes

IP Configuration

Interface

FastEthernet0

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.58.102

Subnet Mask

255.255.255.0

Default Gateway

192.168.58.101

DNS Server

0.0.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

/

Link Local Address

FE80::201:C7FF:FE8E:79AB

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

PC6

Physical

Config

Desktop

Programming

Attributes

IP Configuration

X

Interface

FastEthernet0

IP Configuration

DHCP

Static

IPv4 Address

192.168.58.103

Subnet Mask

255.255.255.0

Default Gateway

192.168.58.101

DNS Server

0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address

/

Link Local Address

FE80::260:2FFF:FE58:6635

Default Gateway

DNS Server

802.1X

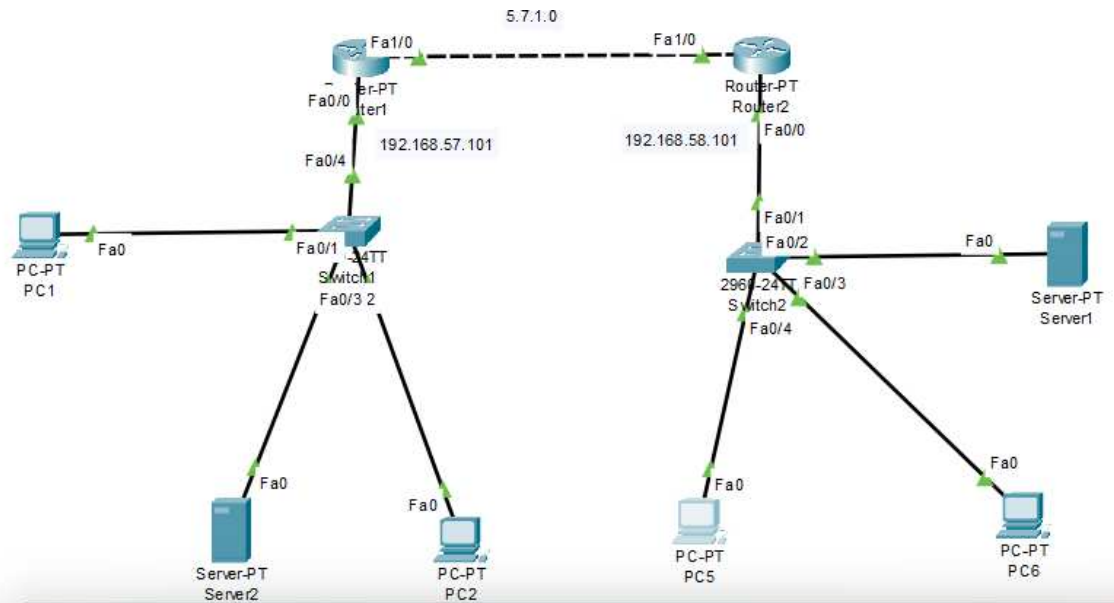
Use 802.1X Security

Authentication

MD5

Username

Password



PC5

Physical Config Desktop Programming Attributes

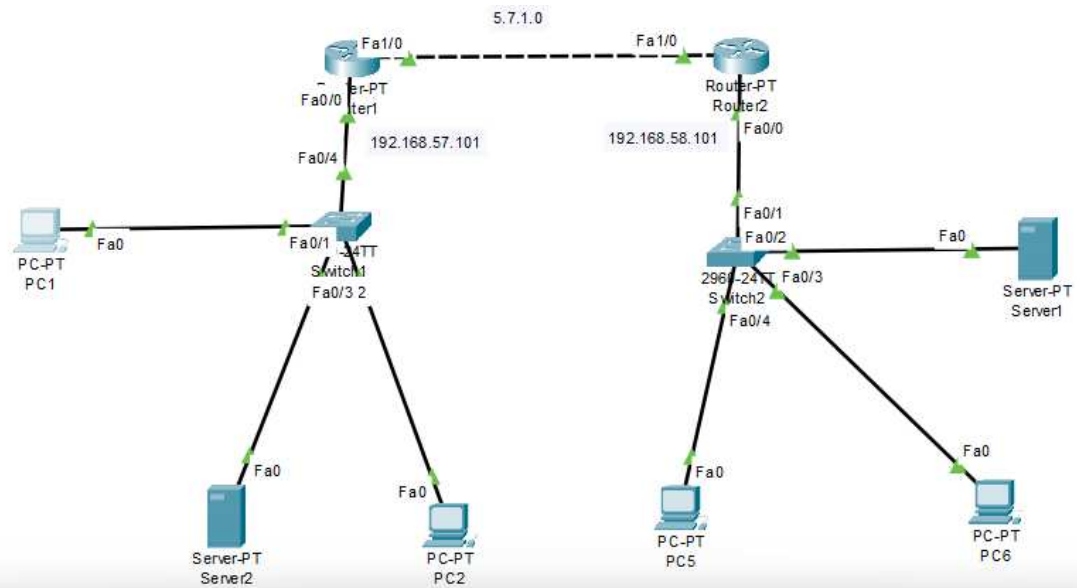
Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.57.103

Pinging 192.168.57.103 with 32 bytes of data:

Reply from 192.168.57.103: bytes=32 time<1ms TTL=126
Reply from 192.168.57.103: bytes=32 time<1ms TTL=126
Reply from 192.168.57.103: bytes=32 time=3ms TTL=126
Reply from 192.168.57.103: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.57.103:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 1ms
```



```
PC1
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.58.103

Pinging 192.168.58.103 with 32 bytes of data:

Reply from 6.0.0.1: bytes=32 time<1ms TTL=126
Reply from 6.0.0.1: bytes=32 time<1ms TTL=126
Reply from 6.0.0.1: bytes=32 time<1ms TTL=126
Reply from 6.0.0.1: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.58.103:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Time: 00:52

Server2

PhysicalConfigServicesDesktopProgrammingAttributes

IP Configuration

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.57.104

Subnet Mask

255.255.255.0

Default Gateway

192.168.57.101

DNS Server

0.0.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

/

Link Local Address

FE80::204:9AFF:FEDE:C18E

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

Server1

PhysicalConfigServicesDesktopProgrammingAttributes

IP Configuration

IP Configuration

☐ DHCP

☒ Static

IPv4 Address192.168.58.104

Subnet Mask255.255.255.0

Default Gateway192.168.58.101

DNS Server0.0.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address/

Link Local AddressFE80::202:4AFF:FED6:8987

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

AuthenticationMD5

Username

Password

Server1

PhysicalConfigServicesDesktopProgrammingAttributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

EMAIL

SMTP Service

☒ ON

☐ OFF

POP3 Service

☒ ON

☐ OFF

Domain Name: gmail.com

Set

User Setup

User

Password

muzammil

muhammad

PC1

Physical Config **Desktop** Programming Attributes

Configure Mail X

User Information

Your Name: muzammil

Email Address: muzammil@hotmail.com

Server Information

Incoming Mail Server: 192.168.58.104

Outgoing Mail Server: 192.168.58.104

Logon Information

User Name: hassaan

Password: ...

Save Remove Clear Reset

PC5

Physical Config **Desktop** Programming Attributes

Configure Mail X

User Information

Your Name: muhammad

Email Address: muhammad@gmail.com

Server Information

Incoming Mail Server: 192.168.58.104

Outgoing Mail Server: 192.168.58.104

Logon Information

User Name: muhammad

Password: ...

Save Remove Clear Reset

